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Industrial Internship Report on

"Banking Information System"

Prepared by

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Executive Summary
This report presents a detailed overview of the industrial internship completed through upskill Campus in collaboration with The IoT Academy and UniConverge Technologies Pvt Ltd (UCT). The internship was conducted over a period of six weeks and focused on developing practical knowledge through a project-based learning approach. The assigned project, "Banking Information System," was implemented using Python programming. The main objective of this project was to simulate fundamental banking operations such as Account creation Login system, PIN security, Deposit / Withdraw, Transaction history, Mini statement in a simple and structured manner. Throughout the internship, emphasis was placed on understanding real-world problem-solving techniques and applying programming concepts effectively. This experience significantly enhanced my analytical thinking, coding ability, and confidence in handling real-time scenarios. The internship also provided valuable exposure to industry practices and improved my overall technical and professional skills.

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1 Preface

The project assigned during this internship is 'Banking Information System'. The main objective of this project is to develop a simple application using Python that performs basic banking operations such as "The system is enhanced with multi-user support and database storage using CSV files managed through the pandas library."account creation, deposit, withdrawal, and balance checking. This project helps in understanding how real-world banking systems function in a simplified manner."

The internship provided by upskill Campus in collaboration with UniConverge Technologies Pvt Ltd gave me an excellent opportunity to gain practical knowledge and work on a real-time project. It helped me understand industry-level expectations and improve my technical and problem-solving skills.

The internship program was well-structured and planned over a duration of six weeks. Initially, basic concepts and project guidelines were provided. Later, the project development phase was carried out step by step, including design, implementation, and testing. Regular guidance and support were provided throughout the internship to ensure successful completion of the project.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

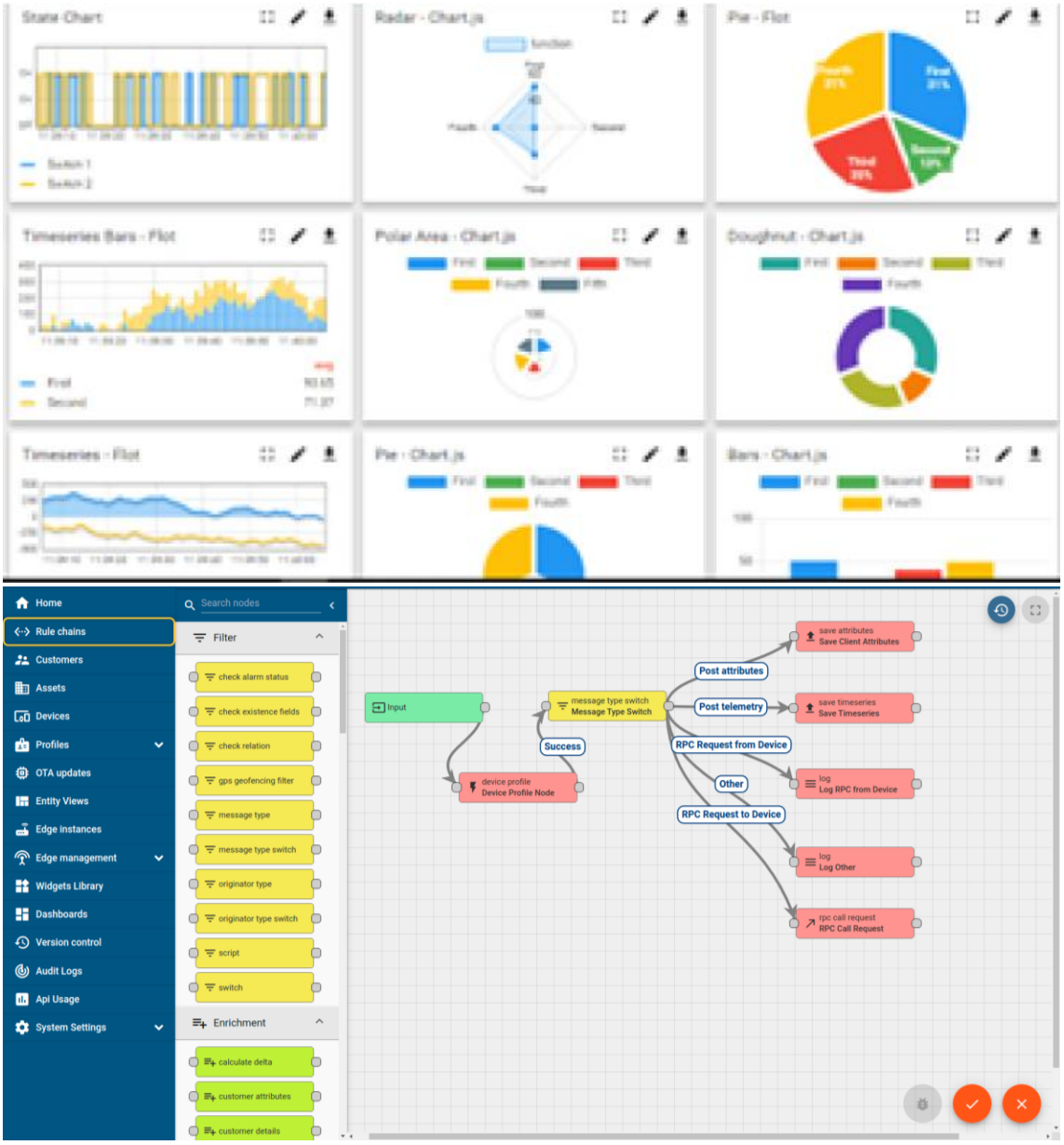
i. UCT IoT Platform (UCT Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

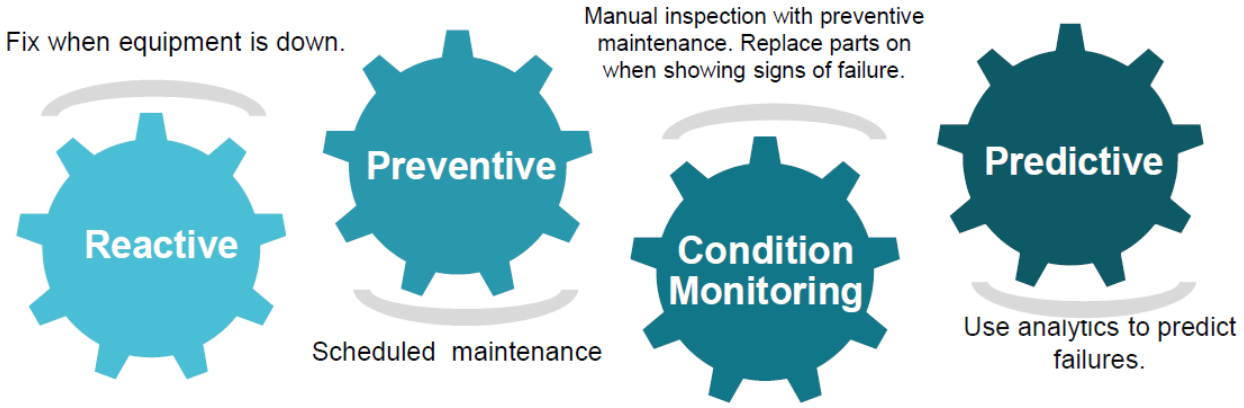
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

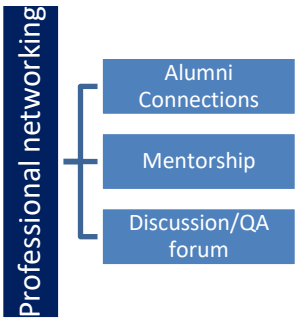
upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

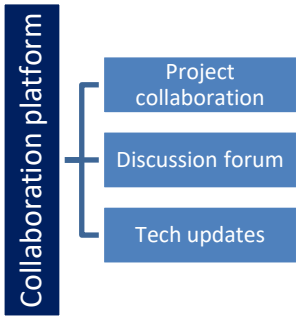
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

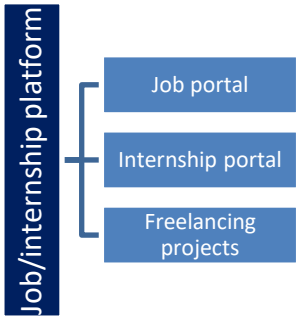
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Python Software Foundation, *Python Documentation*, Available at: https://docs.python.org/

[2] W3Schools, *Python Tutorial*, Available at: https://www.w3schools.com/python/

[3] GeeksforGeeks, *Python Programming Language*, Available at: https://www.geeksforgeeks.org/python-programming-language/

2.6 Glossary

Terms	Acronym
Python	Programming language used to develop the system
Account	A record that stores user details and balance
Deposit	Adding money into the account
Withdrawal	Taking money out of the account
Balance	Current amount of money in the account
Function	A block of code that performs a specific operation
Variable	A container used to store data values
Loop	A control structure used to repeat a set of instructions
Input	Data entered by the user
Output	Result displayed to the user

3 Problem Statement

The objective of this project is to develop a Banking Information System using Python that can perform basic banking operations in a simple and efficient manner. In real-world scenarios, banking systems are complex and require advanced infrastructure; however, this project focuses on creating a simplified version to understand the core functionalities. “Traditional basic programs do not support multiple users or data storage. This project solves this limitation by introducing a database-like system using CSV files, allowing multiple users to create accounts and perform transactions securely.

The system should allow users to create an account, deposit money, withdraw money, and check their account balance. It should also handle user inputs properly and ensure that operations like withdrawal are performed only when sufficient balance is available.

The main challenge of this project is to design a user-friendly and reliable system using basic programming concepts such as functions, loops, and conditional statements. This project helps in understanding how real-world banking applications work in a simplified environment.

Another important aspect of this project is to ensure that the system handles user interactions effectively and maintains accuracy in all operations. The system should be able to respond to different user inputs, including invalid or unexpected entries, and guide the user accordingly. It should also ensure that financial operations are performed correctly without any logical errors. By addressing these aspects, the project aims to build a reliable and user-friendly application that demonstrates the importance of accuracy, validation, and structured programming in developing real-world systems.”

4 Existing and Proposed solution

Existing solutions for banking systems are generally complex and designed for large-scale operations. These systems use advanced technologies such as secure databases, authentication mechanisms, and real-time transaction processing. While they are highly efficient and secure, they are not suitable for beginners as they require deep technical knowledge and infrastructure. Additionally, many simple models available for learning do not provide proper user interaction and lack error handling features, such as checking for insufficient balance or invalid inputs. This limits their effectiveness in demonstrating core banking functionalities.

To overcome these limitations, the proposed solution is a **Banking Information System developed using Python**, which focuses on simplicity and ease of use. The system is designed as a menu-driven application that allows users to perform basic operations such as account creation, deposit, withdrawal, and balance checking. It uses fundamental programming concepts like functions, loops, and conditional statements to ensure clarity and structured execution.

The value addition of this project lies in its simplicity, user-friendly design, and effective demonstration of real-world banking operations. It provides proper input validation, prevents invalid transactions, and helps users understand the internal working of a banking system. This makes it highly useful for learning purposes and serves as a strong foundation for developing more advanced applications in the future.

4.1 Code submission (Github link)

4.2 Report submission (Github link) : first make placeholder, copy the link.

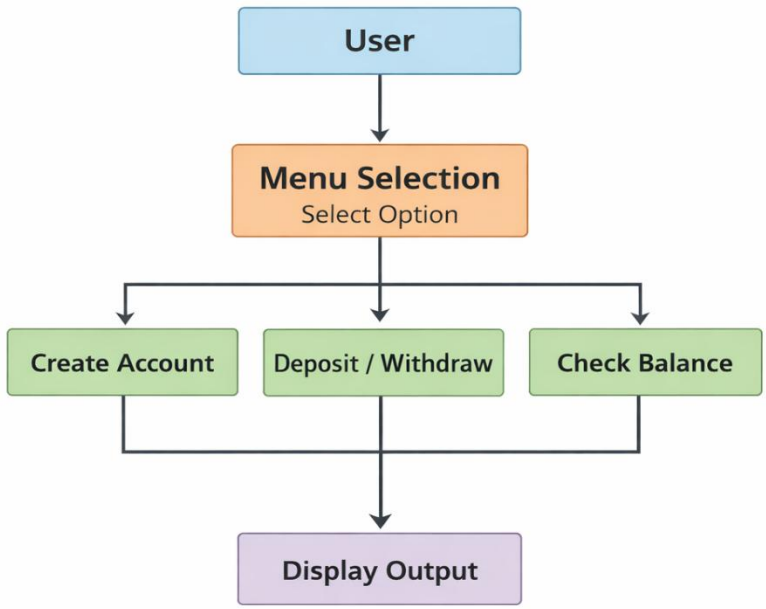
5 Proposed Design/ Model

The proposed Banking Information System is designed using a structured and step-by-step approach to ensure simplicity and efficiency. The system follows a menu-driven flow where the user interacts with the application through different options such as account creation, deposit, withdrawal, and balance inquiry.

The design begins with user input, where the user selects an operation from the menu. Based on the selected option, the system processes the request using appropriate functions. During processing, necessary validations are performed, such as checking for valid input and ensuring sufficient balance for withdrawal operations. Finally, the system displays the output to the user.

The overall design ensures smooth interaction between the user and the system while maintaining accuracy and reliability in all operations. This structured approach helps in better understanding and easy implementation of the system.

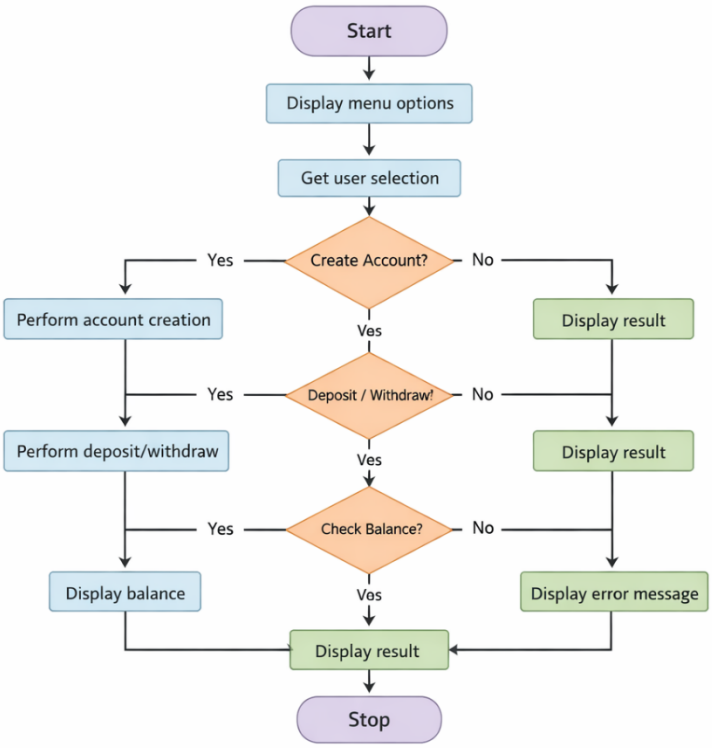
5.1 High Level Diagram (if applicable)



High-Level Diagram of Banking Information System

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

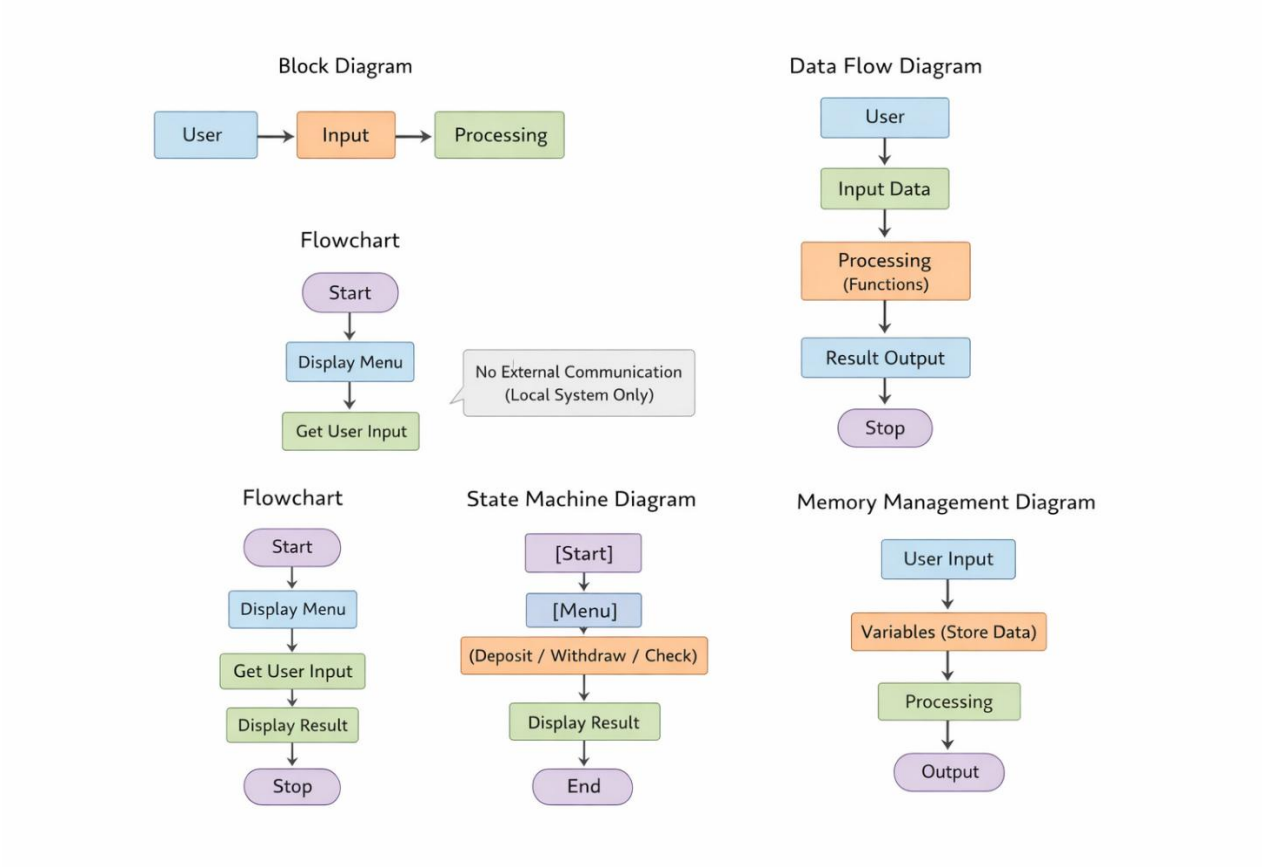


Low-Level Flowchart of Banking Information System

5.3 Interfaces (if applicable)

The Banking Information System uses a Command Line Interface (CLI) for interaction between the user and the system. The user provides input through the keyboard, and the system displays the output on the screen.

The data flow in the system is straightforward. The user provides input, which is processed by the program logic using functions and conditional statements. After processing, the result is displayed back to the user. This ensures smooth communication between the user and the system.



6 Performance Test

The performance testing of the Banking Information System was conducted to evaluate the efficiency, reliability, and correctness of the system under different conditions. The system was tested based on various constraints such as execution time, data handling capability, and accuracy of transactions. Since the system uses a CSV file managed through the **pandas** library, special focus was given to file operations and multi-user data handling.

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Output	Result
Account Creation	Valid name, PIN, balance	Account stored in CSV file	Pass
Duplicate Account	Same name again	Error message displayed	Pass
Login	Correct credentials	Login successful	Pass
Deposit	Valid amount	Balance increases correctly	Pass
Withdraw	Amount ≤ Balance	Balance decreases correctly	Pass
Withdraw	Amount > Balance	Insufficient balance message	Pass
Multiple Users	Different accounts	Data stored separately	Pass
Login	Incorrect credentials	Access denied	Pass
Check Balance	-	Correct balance displayed	Pass

6.2 Test Procedure

The testing process was performed by executing the program multiple times with different input scenarios. Initially, account creation was tested to ensure that user details were correctly stored in the CSV file. Login functionality was verified using both valid and invalid credentials to ensure proper authentication.

Further, banking operations such as deposit, withdrawal, and balance inquiry were tested with different values, including edge cases like insufficient balance and invalid inputs. The system was also tested with multiple users to verify proper data separation and storage.

6.3 Performance Outcome

The performance results indicate that the system operates efficiently for small to medium datasets. The use of pandas for CSV file handling ensures quick data access and modification. The execution time for operations such as deposit and withdrawal was minimal, providing a smooth user experience.

Memory usage remained low due to the lightweight design of the system. The system successfully handled multiple user records without any performance degradation. Accuracy of transactions was maintained in all test cases.

Although the system performs well in its current scope, performance may be affected when handling very large datasets. This can be improved in the future by integrating a dedicated database system such as SQL.

7 My learnings

During this internship project, I gained practical knowledge in developing a real-world application using Python. I learned how to design and implement a Banking Information System by applying core programming concepts such as functions, loops, and conditional statements in a structured manner.

One of the key learnings was working with the pandas library to handle data efficiently. I understood how to use CSV files as a lightweight database to store and manage multiple user records. This helped me learn about data persistence and how real systems maintain user information.

I also gained experience in implementing user authentication using login credentials such as name and PIN. Additionally, I improved my skills in input validation and error handling, ensuring that the system behaves correctly under different conditions.

Through this project, I developed problem-solving skills and learned how to debug errors effectively. I also understood the importance of organizing code in a modular way for better readability and maintenance.

Overall, this internship enhanced my technical skills and gave me confidence to work on real-world projects. It provided a strong foundation for my future career in software development, especially in areas related to application development and data handling.

8 Future work scope

The current Banking Information System is a basic implementation designed to demonstrate core banking functionalities. However, there are several enhancements that can be implemented in the future to make the system more robust, secure, and closer to real-world applications.

One of the major improvements would be replacing the CSV-based storage with a database management system such as MySQL or MongoDB, which can efficiently handle large volumes of data and support concurrent users. This will significantly improve scalability and performance.

Another important enhancement is the development of a Graphical User Interface (GUI) or a web-based application to improve user interaction and accessibility. Security can be further strengthened by implementing encryption techniques for storing PINs and sensitive data, along with multi-factor authentication.

The system can also be extended to include advanced banking features such as fund transfer between accounts, transaction history stored in a database, loan management, and interest calculation. Integration with cloud platforms can further improve availability and reliability of the system.

Additionally, performance optimization techniques and real-time data processing can be implemented to handle larger datasets efficiently. These improvements will transform the current system into a fully functional, industry-level banking application.